LACHOO

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PROJECT REPORT ON

FOOD ORDERING AND BILLING SYSTEM

Course Name: Python Programming Lab

Course Code: CCBCA321

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ABSTRACT

The Food Ordering and Billing System is a Python-based desktop application designed to simplify food ordering and bill generation. It allows the user to enter customer details, select food items, specify quantities, add customizations such as extra cheese, sauce and toppings, modify an order and generate a bill.

The project demonstrates Python programming concepts in a real-world application, including graphical user interface programming, data structures, functions, conditional statements, loops, exception handling and database management.

Python and Tkinter are used for the graphical interface, while SQLite stores completed orders. The qrcode and Pillow libraries support a demonstration QR-payment screen. This QR feature is for demonstration only and does not process real payments.

The final outcome is a desktop food ordering system in which orders can be created, modified, billed and stored for later viewing through Order History.

1. INTRODUCTION

1.1 Background of the Problem

Manual food ordering can lead to calculation mistakes and disorganized records when customers select several items, specify quantities and request customizations. A computerized system can maintain the current order, calculate prices automatically and store completed orders.

1.2 Motivation

This project was selected because food ordering and billing is a common real-world activity that demonstrates multiple Python concepts together: GUI development, functions, data structures, validation, billing logic and database storage.

1.3 Project Requirements / User Needs

• Enter the customer's name.

• Select food items and quantities.

• Add extra cheese, sauce or toppings.

• Add multiple items to one order.

• Increase/decrease quantities or remove items before billing.

• Generate a bill and calculate subtotal, tax and total.

• Select a payment method and save completed orders.

• View previously stored orders through Order History.

1.4 Problem Statement

The objective is to develop a computerized food ordering and billing system that reduces manual calculation and provides an organized method for creating, modifying, billing and storing orders.

1.5 Objectives

• Develop a user-friendly food ordering interface.

• Support food selection, quantities and customizations.

• Calculate item costs, subtotal, 5% tax and total automatically.

• Allow order modification and item removal before billing.

• Provide payment method options and store completed orders in SQLite.

• Display previous orders and demonstrate Python concepts in a functional application.

2. SYSTEM REQUIREMENTS

2.1 Hardware Requirements

• Computer or laptop

• Minimum 4 GB RAM recommended

• At least 500 MB available storage

• Keyboard, mouse and display monitor

2.2 Software Requirements

• Operating system: Windows / Linux / macOS

• Python 3.x

• Python IDLE or Visual Studio Code (mention the IDE actually used)

• SQLite database support

2.3 Technologies Used

Technology	Purpose
Python	Main programming language and application logic
Tkinter	Desktop graphical user interface
SQLite3	Store and retrieve completed order records
qrcode	Generate demonstration QR code
Pillow	Open and display the QR-code image

Note: Confirm the Python version, IDE and libraries against the actual project setup before submission.

3. METHODOLOGY / PROJECT DESIGN

3.1 Project Flow

Start → Enter Customer Name → Select Food Item → Enter Quantity → Select Extra Items → Add Item to Order → Modify / Increase / Decrease / Remove Items → Generate Bill → Calculate Subtotal + 5% Tax + Total → Select Payment Method → Save Completed Order in SQLite → Display Order Confirmation → View Order History if required → End.

3.2 Modules Used

Input Module

Collects the customer name, selected food item, quantity and extra customization options through the Tkinter interface.

Processing Module

Calculates item prices, quantities, extra charges, subtotal, 5% tax and final total. It also handles order modification and item removal.

Output Module

Displays the current order, generated bill, order confirmation and order information through the graphical interface.

Database Module

SQLite stores completed orders. The described records include order ID, customer name, selected items, payment method and total amount.

GUI Module

Tkinter creates labels, entry fields, buttons, dropdown menus, checkboxes, text areas, windows and scrollable sections.

3.3 Python Concepts Used

• Variables store customer name, quantity, price, subtotal, tax and total.

• Dictionaries organize menu items, prices, extras and order information.

• Lists store multiple extra options selected for a food item.

• Conditional statements (if/else) support validation and decision-making.

• Loops process each food item and calculate costs.

• Functions separate operations such as adding items, calculating bills, removing items, saving orders and showing history.

• try-except can handle invalid numerical input.

• int() converts quantity input to an integer.

• SQLite database operations create, insert and retrieve order records.

3.4 Libraries / Frameworks Used

Python: application logic. **Tkinter:** desktop GUI. **sqlite3:** database connection. **qrcode:** demonstration QR generation. **Pillow:** opens and displays the QR image.

4. RESULT AND ANALYSIS

4.1 Objectives Achieved

The application demonstrates order creation, customization, quantity modification, item removal, bill generation and storage of completed orders.

4.2 System Performance

Food selection and order modification are handled through the graphical interface, while billing calculations are performed automatically. SQLite can preserve completed orders after the application closes, provided database operations are implemented correctly.

4.3 Key Observations

The project combines multiple Python concepts in one application. Functions separate operations, lists and dictionaries organize menu/order information, and validation improves reliability.

4.4 Features Shown in Screenshots

• Customer name and food menu selection.

• Quantity and extra cheese, sauce and toppings options.

• Current order with increase, decrease and remove controls.

• Generated bill showing item cost, extras, subtotal and 5% tax.

• Cash, card and UPI/QR payment method selection.

• QR screen labelled as demonstration only, with no real payment processing.

• Order History showing customer, items, payment method and total.

4.5 Limitations

• QR payment is a demonstration and does not process real payments.

• Menu items and prices are predefined.

• The system is a desktop application, not an online food-ordering platform.

• Online delivery and live order tracking are not included.

• User login and authentication are not included.

5. CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The Food Ordering and Billing System demonstrates a Python-based desktop application for selecting food, customizing orders, modifying quantities, generating bills and storing completed orders. Tkinter makes the application interactive, while SQLite supports storage of order records.

The project demonstrates functions, dictionaries, lists, loops, conditional statements, exception handling, type conversion and database connectivity. It is a practical example of using Python to address a real-world problem.

5.2 Future Scope

• Add online food-ordering functionality.

• Introduce user login and authentication.

• Create an admin panel to update menu items.

• Integrate a real payment gateway if required and implemented securely.

• Add order tracking and customer notifications.

• Generate sales reports and graphical analysis.

• Add search/filtering for previous orders.

• Explore cloud database storage and improve UI design.

6. REFERENCES

The following official documentation sources are relevant. Include only sources actually consulted during development.

• Python Documentation — https://docs.python.org/3/

• Tkinter Documentation — https://docs.python.org/3/library/tkinter.html

• sqlite3 Documentation — https://docs.python.org/3/library/sqlite3.html

• SQLite Documentation — https://www.sqlite.org/docs.html

• Pillow Documentation — https://pillow.readthedocs.io/

• qrcode package — https://pypi.org/project/qrcode/

7. APPENDIX

7.1 Source Code and Project Files

Include the complete Python source code and supporting project files. Use main.py as the main filename only if that is the actual filename in your project.

The SQLite database may be named food_orders.db and the generated QR image may be named demo_qr.png, if these are the actual filenames used.

7.2 GitHub Repository (Mandatory)

GitHub Repository Link: _____

Upload the complete project to GitHub and replace this blank line with the actual repository URL.

7.3 Execution / Output Screenshots

The following pages contain the screenshots provided for the project, each with a figure caption.

APPENDIX: PROJECT OUTPUT

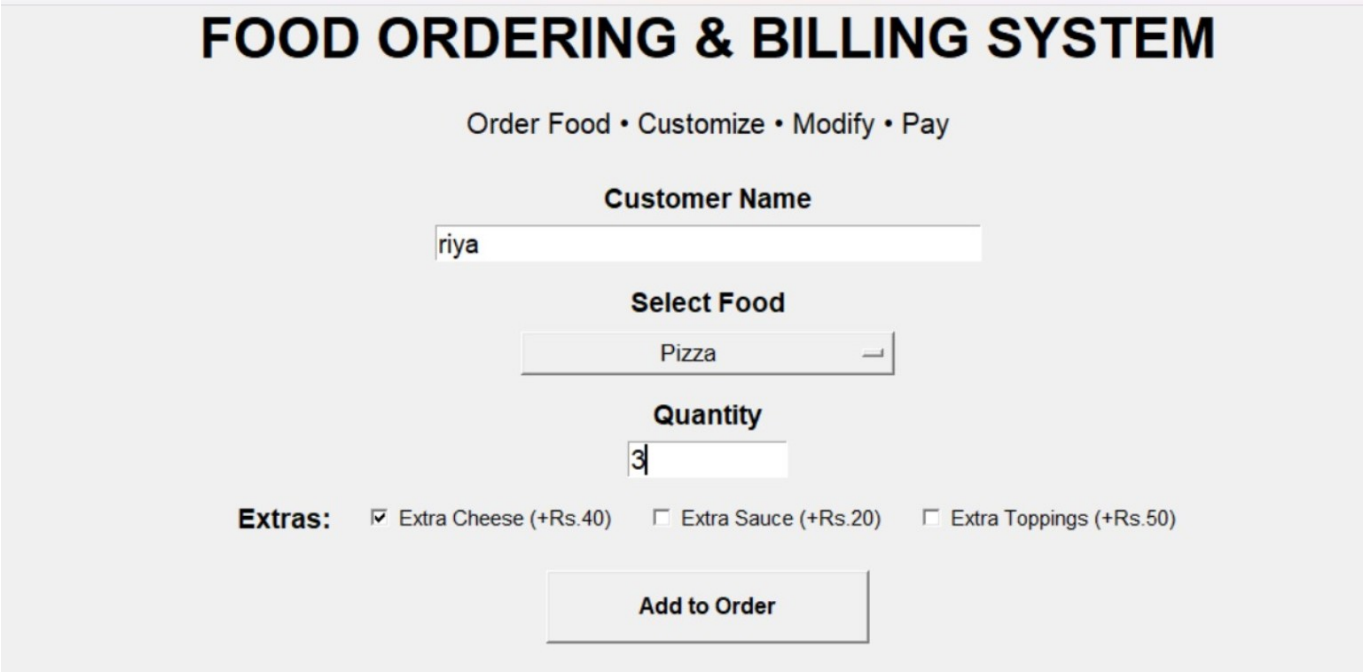


Figure 1: Main Food Ordering Interface

APPENDIX: PROJECT OUTPUT



Figure 2: Current Order and Modification Options

APPENDIX: PROJECT OUTPUT

Generated Bill

```
========================================
             FOOD BILL
========================================
Customer: riya
----------------------------------------
Pizza                 x 3   Rs. 750.00
  + Extra Cheese        Rs. 120.00
----------------------------------------
Subtotal              : Rs. 870.00
Tax (5%)              : Rs. 43.50
```

Figure 3: Generated Bill

APPENDIX: PROJECT OUTPUT

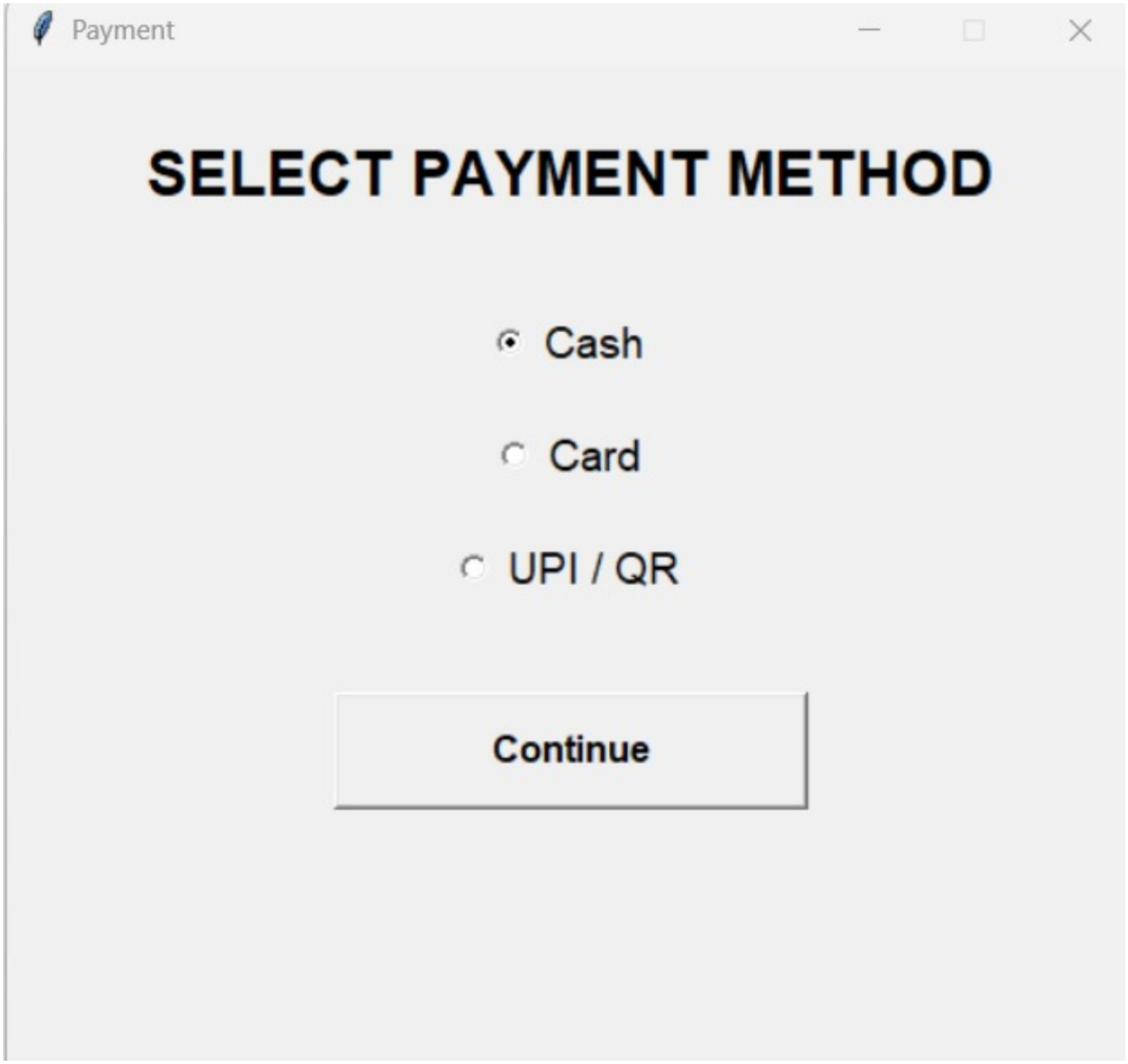


Figure 4: Payment Method Selection

APPENDIX: PROJECT OUTPUT



Figure 5: Demonstration QR Payment Window

APPENDIX: PROJECT OUTPUT



Figure 6: Order History